

Solution Requirements

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can access the Voice-Based Concept Understanding Analyser directly through the web interface without requiring login or account registration, making the application simple and easily accessible.</i>	low
2	Authorization levels	<i>The current version supports a single user role (student/learner). All users have access to concept selection, voice recording, answer submission, and feedback generation. No administrator module is included</i>	low

3	External interfaces	<i>The system interacts with the browser's Web Speech API for speech-to-text conversion and uses locally stored JSON knowledge bases along with NLP libraries for concept evaluation. No external AI APIs are required during analysis.</i>	medium
4	Transactions processing	<i>The user selects or enters a concept, records or enters an explanation, submits the response, the Flask backend processes the text using NLP techniques, evaluates conceptual understanding, and returns the mastery score, detected misconceptions, missing concepts, and personalized feedback.</i>	High
5	Reporting	<i>The system generates an analysis report displaying the transcript, mastery score, concept coverage, identified misconceptions, missing concepts, and personalized learning feedback.</i>	High
6	Business rules, etc.	<i>User responses should contain meaningful explanations for accurate evaluation. The system compares the submitted explanation with predefined concept knowledge bases and calculates similarity scores before generating feedback.</i>	Medium
7	Compliance to laws or regulations	<i>The application does not collect or permanently store personal or sensitive user information. Voice recordings are processed only for concept evaluation, minimizing privacy concerns</i>	low

8	Other	The system should support adding new concepts or updating the knowledge base without requiring major modifications to the application architecture.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	The system should analyze the submitted explanation and generate feedback quickly to provide a smooth user experience.	Analysis results generated within 2–5 seconds for normal responses.
2	Scalability	The application should support adding more concepts, larger knowledge bases, and advanced NLP models without significant architectural changes.	Modular architecture allows future deployment on cloud platforms and integration of additional AI models.
3	Security & Data Privacy	The system should protect user inputs and process responses securely without permanently storing voice recordings or sensitive information.	Input validation implemented; all processing performed server-side without long-term storage of user data.
4	Reliability & Availability	The application should provide consistent concept evaluation with minimal failures during normal usage.	Reliable operation during local deployment with target 99% availability after cloud deployment.
5	Usability & Accessibility	The interface should be simple and intuitive so students can easily record responses, submit explanations, and understand the generated feedback without prior training.	Users can complete concept evaluation in a few simple steps with clear instructions and responsive interface.

6	Maintainability	<i>The application should have a modular code structure separating frontend, backend, NLP engine, and knowledge base to simplify future enhancements and maintenance.</i>	Well-organized project structure with separate modules (app.py, nlp_engine.py, templates, static files, and JSON knowledge base).
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